

RAMTIN SOLEYMANI

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EDUCATION

Toronto Metropolitan University (Formerly Known as Ryerson)

Sep. 2021 – June 2026

Bachelor of Engineering, Electrical Engineering (Co-op)

Toronto, ON

- **Final-Year GPA:** 3.67/4.33
- **Relevant Coursework:** Computer Architecture, SoC Design, Digital Systems, Embedded Systems

TECHNICAL SKILLS

Languages & Technologies: SystemVerilog, Verilog, C/C++, Python, Bash/Shell, Tcl, Synopsys VCS, Vivado, Quartus II
Hardware & Interfaces: Xilinx Artix-7, Altera Cyclone IV, ARM Cortex-M3, GPU Architecture, AXI/AHB, UART, I2C/SCCB
Verification: UVM, Self-Checking Testbenches, Scoreboards, SVA, Functional Coverage, Waveform Debug
Digital Design: RTL Design, CDC, STA, Synthesis, Timing Closure, Memory Hierarchy, Cache Coherence

EXPERIENCE

Electrical Engineering Intern

Aug. 2024 – Aug. 2025

Hydro One Networks Inc. | Ontario Electricity Transmission Utility

Toronto, ON

- Diagnosed recurring 115 kV & 230 kV transmission outages by comparing outage history and weather data, identifying **lightning backflash** and supporting a targeted grounding solution with a Senior Engineer.
- Verified construction safety near 500 kV transmission lines by re-modelling the design in PLS-CADD and analyzing **60 Hz EMF** and **induced current** results in **EPRI TLW-Gen2** against IEEE & Hydro One safety standards.
- Eliminated manual report preparation by **automating** project status tracking with **Microsoft Power Automate**.
- Authored a **20+ page report** using IEEE/company standards on HV solar-farm risks, leading to project cancellation.

Electrical Design Intern (Part-Time)

May 2023 – Aug. 2024

R.J. Burnside & Associates Limited | Engineering Consultancy

Pickering, ON

- Produced **10+ electrical drawings** in **AutoCAD** while working part-time during full-time engineering studies.
- Decreased projected costs by **up to 40%** across **5+ street-lighting designs** by optimizing layouts and equipment.
- Prevented submission errors through **technical peer reviews** of completed electrical design drawings.

PROJECTS

FPGA Camera / Display Pipeline | SystemVerilog, UVM, CDC

GitHub 

- Built a 320x240 resolution OV7670-to-VGA **SystemVerilog** pipeline with SCCB control, scaling output to 640x480.
- Stabilized camera-to-VGA pixels across **2 clock domains** by adding an async FIFO with Gray-coded CDC pointers.
- Added FIFO buffering to the framebuffer, reducing frame-tear severity by **32.91%** across **25 frames** versus the baseline, using a **UVM scoreboard**.
- Enabled edge detection and PC snapshots by adding a **3x3 Sobel design** and UART export of **320x240** frames.

32-bit Multi-Cycle CPU | SystemVerilog, Directed Tests, Waveform Debug

GitHub 

- Implemented a **32-bit SystemVerilog CPU** on Basys 3 and DE2-115 with a Harvard architecture and 19-state FSM.
- Achieved **100 MHz timing closure** by adding one FSM cycle, improving WNS from **-0.742 ns** to **+3.275 ns**.
- Optimized **resource use** by **about 30%** through datapath reuse and verified **30+ instructions** using Quartus II waveform simulation.

Network-on-Chip Interconnect | SystemC, Self-Checking Tests, Linux

GitHub 

- Created a **4x4 SystemC NoC** with 16 wormhole routers and mesh/torus topology support.
- Validated 384 packets and 1,920 flits with **zero routing errors** or deadlocks using automated self-checking tests.
- Lowered uniform-traffic latency by **1.9%**, from **25.62 ns** to **25.14 ns**, by adding torus wraparound links.

Embedded Media Center | C/C++, OOP, Firmware

GitHub 

- Integrated LCD graphics, audio, joystick input, menus, and games on an ARM Cortex-M3 using **C & C++** firmware.
- Cut firmware memory usage by **60%** by refactoring menus and game logic into reusable **OOP modules**.

ML-Based RTL Timing Closure Predictor | Python, Tcl, Vivado

May 2026 – Present

- Automated Vivado **synthesis** and **implementation** with **Python/Tcl** to extract validated per-clock timing labels.
- Developing an **RTL-only ML** workflow to estimate WNS, FMax, and timing risk before full FPGA implementation.

Open to relocation.

Canadian Citizen.